

HW 5

46-881, Section M

Submission

Please submit a PDF or a notebook file with your answers.

Description

Sensors_M.csv contains data from an imaginary STech company that manufactures industrial IoT sensors for smart factories and energy infrastructure.

STech operates three factories (F01, F02, F03), each producing two main sensor types: TempSensor — measures temperature in harsh environments, and MotionSensor — detects vibrations and movement in equipment.

Each factory runs three daily shifts (A, B, and C).

Goal: Identify the underperforming factories and shifts.

Questions

1. (10 points) Load the dataset into the variable “sensors”.
2. (10 points) Check the summary statistics. Any data errors?
3. (20 points) Calculate the total output for each factory.
4. (20 points) Calculate the total output by factory and sensor type.
5. (5 points) Which shift has the highest average output?
6. (10 points) Identify the top 3 days with the highest total defect counts across all factories.
7. (15 points) Add another variable to the dataset: DefectRate. Defect Rate is the ratio of Defective Units to Units Produced.
8. (5 points) Find the day-shift-factory combinations where the defect rate exceeded 10%. How frequent are these cases?
9. (5 points) Make graphs.
 - a) A line plot of daily units produced per factory. All three factories on one graph. Read about possible parameters for lineplot.
 - b) A bar chart of total units produced by shift and factory. Read about possible parameters for barplot.
10. (10 points) You now have two different ways to describe a factory's overall defect rate:
 - (a) the average of the individual DefectRate values from Question 7, and
 - (b) the ratio of total defective units to total units produced (summing first, then dividing).Compute both, by factory. Do they match? If not, why not, and which one is the more defensible way to describe a factory's overall defect rate?
11. (10 points) Synthesis: based on everything above, output, defect rate, and downtime, which factory and which shift would you flag as underperforming? Justify your answer using more than one metric.